LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING

NASA JPL ION-DTN CVE-2026-123999 Network Management Daemon Daily Priority

CVE / Vulnerability Threat Intelligence Report

Classification	TLP:CLEAR
Published	2026-11-10
Version	1.0
Author	Lost Boys Cyber
Report Type	CVE / Vulnerability Threat Intelligence Report

TLP:CLEAR | Lost Boys Cyber | NASA JPL ION-DTN CVE-2026-123999 Network Management Daemon Daily Priority - CVE / Vulnerability Threat Intelligence Report

Published: 2026-11-10 | TLP:CLEAR | Version: 1.0 | Author: Lost Boys Cyber

AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

Table of Contents

1. Executive Summary
2. Vulnerability Metadata
3. Source Review & Confidence
4. Vulnerability Details
5. Aviation and Aerospace Sector Impact
6. Attack Scenarios
7. Affected Products and Exposure
8. Mitigation and Workarounds
9. Detection Engineering
    9.1. Sigma Detection Rule
    9.2. Hunt Query
10. Threat Hunting
    10.1. Hunt Hypothesis
11. MITRE ATT&CK Mapping
12. Validation / Lab Testing
13. Recommended Actions Summary
14. References

NASA JPL ION-DTN CVE-2026-123999 Network Management Daemon Daily Priority

1. Executive Summary

CVE-2026-123999 is a newly published high-severity availability vulnerability in NASA JPL ION-DTN versions 3.6.2 through 4.2.0. The published description identifies a heap-based buffer overflow in the Network Management daemon, `nm_mgr`, when deeply nested AMP payloads reach fixed-size string-formatting buffers through unbounded concatenation. A malicious or misconfigured agent can reportedly send oversized AMP reports, tables, or macros over the UDP CLA/AMP path, or trigger the flaw through interactive UI rendering, causing `nm_mgr` to crash and interrupt telemetry collection and operator visibility.

This is an aerospace/space-communications prioritization item because Delay/Disruption-Tolerant Networking (DTN) deployments can be used in constrained, remote, or intermittently connected environments. The reviewed sources support a denial-of-service concern, not a claim of remote code execution, compromise of spacecraft flight systems, active exploitation, or a broadly exposed aviation impact. Operators should inventory ION-DTN use, identify reachable management and AMP paths, apply the upstream fix, and validate management-plane resilience.

2. Vulnerability Metadata

Field	Value
CVE	CVE-2026-123999
Product / component	NASA JPL ION-DTN Network Management daemon (`nm_mgr`)
Affected versions	3.6.2 through 4.2.0
Fixed release reference	Upstream release `ion-open-source-4.2.1-a.2` referenced by the advisory
CVSS	7.5 (CVSS v3.1); 8.7 (CVSS v4.0 secondary score)
Primary impact	Availability / loss of management telemetry and operator visibility
Exploitation status	No active-exploitation statement found in reviewed sources

3. Source Review & Confidence

Source	Contribution	Confidence
NVD CVE API record	Published description, affected range, metrics, source links	High
GitHub Security Advisory GHSA-gmhw-vmwr-wrxh	Upstream-associated advisory and remediation reference	High
NASA JPL ION-DTN release	Upstream release provenance for 4.2.1-a.2	High
VulnCheck / Rapid7 CVE records	Independent public reproduction of published technical context	Medium

The CVE was published on 2026-11-09. Public material reviewed for this run supports the daemon-crash and visibility-loss risk. It does not establish deployment prevalence, whether UDP CLA/AMP is exposed to untrusted networks, or an operational safety consequence for any particular aviation or space mission.

4. Vulnerability Details

The published record describes unsafe string-formatting behavior in `nm_mgr` when it processes deeply nested AMP content. The named paths include oversized AMP reports, tables, and macros over UDP CLA/AMP, plus local interactive UI rendering. A crash in the network-management function can remove telemetry and operator visibility at the time it is most needed.

Treat this as a management-plane resilience issue. It is especially important to distinguish data-plane DTN traffic from administration/management paths and to identify which agents are trusted to send AMP. Do not infer that all ION-DTN nodes or spacecraft-adjacent systems are externally reachable.

5. Aviation and Aerospace Sector Impact

Sector segment	Operational relevance	Likely exposure path	Defender action
Satellite and ground-segment DTN operations	Loss of management telemetry can complicate fault diagnosis and mission support	UDP CLA/AMP reachable from an untrusted or poorly segmented agent	Patch, restrict source networks, test failover visibility
Aerospace R&D and test ranges	Lab/test deployments may inherit permissive management routing	Shared engineering networks or untracked test agents	Inventory images and isolate test management networks
Aviation communications research	DTN experiments may bridge enterprise, field, or mobile networks	Misconfigured routing or partner-connected nodes	Verify trust boundaries and partner access
MRO / supplier environments	Supplier-managed systems can carry unreviewed software versions	Third-party remote support or unmanaged appliances	Require version attestation and patch evidence

Third-party and supply-chain considerations: obtain an authoritative software bill/version statement from integrators, ground-segment suppliers, and managed research partners. The report does not claim that ION-DTN is deployed in commercial airline operations; sector prioritization follows the product's space-communications relevance.

6. Attack Scenarios

Scenario	Preconditions	Defender focus
Malformed AMP causes daemon crash	Attacker or faulty peer can reach UDP CLA/AMP management path	Segment and authenticate management peers; monitor daemon restarts
Misconfigured internal agent sends oversized data	Trusted agent is faulty or compromised	Validate message-size controls and source allowlists
Local UI rendering crash	Operator or local process renders crafted nested content	Patch management workstations and restrict untrusted inputs

7. Affected Products and Exposure

Exposure pattern	Why it matters	Priority
ION-DTN 3.6.2–4.2.0 with reachable management agent paths	Published network-triggerable DoS condition	High
Ground segment or mission-support node with remote partners	Trust boundaries can expand beyond an operator's core network	High

Isolated test node with no untrusted AMP input	Lower external reachability but still patch during maintenance	Medium
Fixed release with tested monitoring/failover	Residual risk depends on segmentation and historical events	Medium

8. Mitigation and Workarounds

1. Inventory ION-DTN installations, package versions, role, network interfaces, and configured management/AMP peers.
2. Upgrade affected deployments using the upstream remediation release path; validate compatibility in an authorized staging or mission test environment first.
3. Limit UDP CLA/AMP and network-management traffic to explicitly authorized peers using segmentation, host firewalls, and route controls.
4. Monitor `nm_mgr` process exits, restarts, watchdog events, and telemetry gaps; preserve logs before restarts overwrite evidence.
5. Establish message-size and input-validation expectations at the application/integration layer where supported.
6. Confirm redundant monitoring and operational fallback procedures so management visibility survives a daemon failure.

9. Detection Engineering

Signal	Telemetry source	Analytic goal
`nm_mgr` exit or restart	systemd, EDR, application logs	Detect likely impact and availability degradation
UDP traffic to AMP/CLA port from new peer	Network flow / firewall logs	Identify unexpected management-plane source
Burst of oversized or malformed management messages	DTN/agent logs and packet telemetry	Identify triggering conditions

9.1. Sigma Detection Rule

```
title: ION-DTN Network Management Daemon Unexpected Termination
id: 4a762a7d-6767-4b3f-9c10-1d99436f1640
status: experimental
logsource:
  product: linux
  service: systemd
detection:
  selection:
    SYSLOG_IDENTIFIER|contains: 'systemd'
    message|contains|all:
      - 'nm_mgr'
      - 'Main process exited'
  condition: selection
level: high
```

9.2. Hunt Query

```
DeviceProcessEvents
| where Timestamp > ago(30d)
| where FileName in~ ('nm_mgr', 'systemd')
| where ProcessCommandLine has 'nm_mgr' or InitiatingProcessCommandLine has 'nm_mgr'
| project Timestamp, DeviceName, ActionType, FileName, ProcessCommandLine, InitiatingProcessCommandLine
| order by Timestamp desc
```

Tune to local process names, service-unit names, and EDR retention. A restart alone is not proof of exploitation; correlate it with packet data and application error logs.

10. Threat Hunting

10.1. Hunt Hypothesis

If the issue is triggered, `nm_mgr` will restart or terminate near anomalous inbound management-plane traffic, particularly from an unapproved peer or a source with an unusual packet-size distribution.

```
index=network sourcetype=netflow dest_port IN (4556,4557)
| stats count sum(bytes) as bytes max(bytes) as max_bytes values(src_ip) as sources by dest_ip,
dest_port, _time
| where max_bytes > 100000 OR count > 1000
| sort - max_bytes
```

Port values are placeholders and must be replaced with the ports used by the local ION-DTN/AMP configuration. Correlate candidates to daemon restart logs before escalation.

11. MITRE ATT&CK Mapping

Technique	Name	Rationale
T1499	Endpoint Denial of Service	Published impact is a daemon crash / interruption of management visibility
T1190	Exploit Public-Facing Application	Conditional only: relevant if an exposed management endpoint is reachable; not asserted as confirmed exploitation

12. Validation / Lab Testing

No malformed payload was sent and no production system was tested. In an authorized isolated environment, validate the fixed release, service recovery, logging, source restrictions, and telemetry failover. Do not send crafted AMP payloads to operational ground, flight, or safety-related environments.

13. Recommended Actions Summary

Priority	Owner	Action
High	Space/ground systems owner	Locate ION-DTN deployments and confirm version/remediation plan
High	Network security	Restrict management/AMP peer reachability to approved sources
High	SOC / NOC	Alert on `nm_mgr` exits and telemetry gaps; preserve evidence
Medium	Supplier management	Obtain version and patch attestations from integrators and partners

14. References

- NVD, CVE-2026-123999: https://nvd.nist.gov/vuln/detail/CVE-2026-123999
- GitHub Security Advisory, GHSA-gmhw-vmwr-wrxh: https://github.com/nasa-jpl/ION-DTN/security/advisories/GHSA-gmhw-vmwr-wrxh
- NASA JPL ION-DTN release `ion-open-source-4.2.1-a.2`: https://github.com/nasa-jpl/ION-DTN/releases/tag/ion-open-source-4.2.1-a.2
- VulnCheck advisory: https://www.vulncheck.com/advisories/ion-dtn-heap-buffer-overflow-via-nm-mgr-amp-rendering
- Rapid7, CVE-2026-123999: https://www.rapid7.com/db/vulnerabilities/cve-2026-123999/